Unexploitable modal agents that cooperate with themselves: a census

Lindheim, Björn

1 Introduction

Setting

Two programs play a one-shot prisoner's dilemma, and each can read the other's source code. Following Barasz et al. [1], a *modal agent* decides by asking what can be proved about the opponent, written with the provability box $\Box$ of the logic GL: p means "the opponent cooperates with me" and q "the opponent cooperates with DefectBot". FairBot is $\Box p$: it cooperates exactly when it can prove that the opponent cooperates with it. An agent is *unexploitable* if no modal opponent ever ends up defecting while the agent cooperates, and *self-cooperating* if two copies cooperate. We take guarded formulas (every atom under a box) built from p, q, $\bot$, $\neg$, $\wedge$, $\vee$, $\Box$, measure size as the number of nodes, and call two agents behaviourally equal when they make the same moves against every opponent. Agents using only p are the rank-0 agents of Barasz et al.

A LessWrong post [31] asked whether every rank-0 agent that is unexploitable and self-cooperating is FairBot. A comment there shows that the answer is yes up to modal depth 1 and no from depth 2, with examples such as $\Box\Box p$, FairBot over PA+1 and $\Box^k p$.

Result

- **Rank 0.** Among rank-0 agents of size at most 15 there are exactly 5031 classes that are unexploitable and self-cooperating (4374 of them also cooperate with FairBot and CooperateBot). Only $\Box p$ itself is FairBot; every other class is told apart from FairBot by an explicit opponent.
- **With q.** Allowing q, up to size 14 there are 26449 such classes (16756 up to behaviour); 387 of them behave exactly like FairBot against every opponent, and each of the others is told apart from FairBot by an explicit opponent.
- **Why some different formulas are FairBot.** Two agents behave identically against every opponent exactly when they have the same moves against DefectBot and agree on every input the opponent can actually

produce: until the agent first cooperates, the opponent's move against it equals its move against DefectBot. So, for example, $\Box(p \vee (q \wedge \Box\bot))$ is behaviourally FairBot.

- **A small example that is not FairBot.** $\neg\Box\bot \wedge \Box(p \vee \Box\bot)$ (size 9) is unexploitable and cooperates with itself and with CooperateBot, yet mutually defects with FairBot; it is the agent WaitFairBot$_1$ of Barasz et al. Every smaller rank-0 agent that is unexploitable and self-cooperating cooperates with FairBot.

Why it matters

The LessWrong question was answered with a few examples. These counts make the answer quantitative: among small agents, FairBot is one of thousands of unexploitable self-cooperators, and for each one there is a concrete opponent that tells it apart from FairBot. The criterion for behavioural equality explains which syntactically different agents are nevertheless FairBot.

Proved vs checked

The equality criterion and the FairBot-equivalence of $\Box(p\vee(q\wedge\Box\bot))$ are proved in Lean 4, in the height semantics of closed GL fixed points; the link to provability in arithmetic is Solovay's theorem [29], cited, not formalised (see also [12]). The counts are exhaustive computations up to the stated sizes, with every separating opponent recorded and re-solved by the checks. The minimality statement for the size-9 agent is an exhaustive search over smaller rank-0 agents.

Novelty statement

This result extends Barasz et al. (arXiv 1401.5577): modal agents, unexploitability, rank, and WaitFairBot_K; "Is there only one FairBot?" (LessWrong, 2026), with Zac Hatfield-Dodds's comment answering it to modal depth 1 and giving examples. Our prior-art search (2026-10-21) covered:

- Barasz, LaVictoire, Christiano, Fallenstein, Herreshoff, Yudkowsky, "Robust Cooperation in the Prisoner's Dilemma: Program Equilibrium via Provability Logic" (arXiv 1401.5577): unexploitability (Section 2), rank and Theorem 4.10 (Section 4), WaitFairBot_K and the obstacle theorem (Section 5).

- LessWrong, "Is there only one FairBot?" (transhumanist_atom_understander, 29 Aug 2026) and comments: Zac Hatfield-Dodds shows the uniqueness conjecture holds up to modal depth 1 for rank-0 agents and fails from depth 2 (Box Box p; FairBot over PA+1 = Box(p or Box False); Box^k p for k >= 3); PrudentBot separates Box Box p from FairBot; the author notes Box Box p is not provably unexploitable in PA. No enumeration or counts.

- LessWrong / Alignment Forum wiki "Modal combat": DefectBot, CooperateBot, FairBot, PrudentBot, CliqueBot, TrollBot; no uniqueness result, no enumeration.
- LaVictoire et al. 2014 (MIPC); comment thread on "Robust Cooperation in the Prisoner's Dilemma" (LessWrong, on instability of FairBot equilibria); Hobson, "Yet More Modal Combat" (2021); Critch, parametric bounded Löb (arXiv 1602.04184); Critch, Dennis & Russell (arXiv 2208.07006); MIRI provability repository (github.com/machine-intelligence/provability, a modal-agent evaluator, no published census).
- Out of scope by direction: probabilistic blends (Oesterheld arXiv 2211.05057; Cooper et al. arXiv 2412.14570).
- Web searches for deterministic blends, behavioural distance and classifications of modal agents (2026-10-20/21); none means not found in this search.

If you know of related work we missed, please open an issue.
We believe this result is new. Our prior-art search (2026-10-21) covered:

- Barasz, LaVictoire, Christiano, Fallenstein, Herreshoff, Yudkowsky, "Robust Cooperation in the Prisoner's Dilemma: Program Equilibrium via Provability Logic" (arXiv 1401.5577): unexploitability (Section 2), rank and Theorem 4.10 (Section 4), WaitFairBot_K and the obstacle theorem (Section 5).
- LessWrong, "Is there only one FairBot?" (transhumanist_atom_understander, 29 Aug 2026) and comments: Zac Hatfield-Dodds shows the uniqueness conjecture holds up to modal depth 1 for rank-0 agents and fails from depth 2 (Box Box p; FairBot over PA+1 = Box(p or Box False); Box^k p for k >= 3); PrudentBot separates Box Box p from FairBot; the author notes Box Box p is not provably unexploitable in PA. No enumeration or counts.
- LessWrong / Alignment Forum wiki "Modal combat": DefectBot, CooperateBot, FairBot, PrudentBot, CliqueBot, TrollBot; no uniqueness result, no enumeration.
- LaVictoire et al. 2014 (MIPC); comment thread on "Robust Cooperation in the Prisoner's Dilemma" (LessWrong, on instability of FairBot equilibria); Hobson, "Yet More Modal Combat" (2021); Critch, parametric bounded Löb (arXiv 1602.04184); Critch, Dennis & Russell (arXiv 2208.07006); MIRI provability repository (github.com/machine-intelligence/provability, a modal-agent evaluator, no published census).
- Out of scope by direction: probabilistic blends (Oesterheld arXiv 2211.05057; Cooper et al. arXiv 2412.14570).

- Web searches for deterministic blends, behavioural distance and classifications of modal agents (2026-10-20/21); none means not found in this search.

We found no earlier statement of it, but we can't be certain. If you know of prior work, please open an issue.

How it was produced

This result was found by AI research agents (the autoresearch harness). It was partly checked by scripts that run Lean proofs (claims K11, K18) and partly checked by script (claims K51, K53, K2); the proof and check files are included so anyone can re-run them.

2 Claims

Claim K51

In rank-0 L (atom p, guarded, size ≤ 15) exactly 5031 classes are unexploitable and self-cooperating, 4374 of them cooperating with FairBot and CooperateBot; only $\Box p$ is FairBot, every other one is separated from it by an explicit opponent.

Status: derived ů author check only.

Verified.

- `checks/K51-c046_ponly15.py`: guarded p-only class counts per size 1..15; 15092 exploitable self-cooperators each exploited by a re-solved letterless opponent; 5031 unexploitable self-cooperators (per-size list); 5030 non-FB each separated from []p by a constructed re-solved opponent; FB-defecting 4,8,16,33,80,171,345 (sizes 9..15), CB-defecting 2,4,8,16,41,87,181, FB&CB-cooperating 4374; depth-1 reproduction; K1 prefix counts — pass

- `checks/K51-c046_top.py`: K51 engine enumerated to size 13 with and without primitive top (size 1): guarded counts differ only at sizes 1,2; unexploitable self-cooperator counts per size identical (1202); top itself exploitable — pass

Evidence.

- reasoning: Census language is bot,p,~,&,\|,box with top = ~bot (size 2). Top is absorbed in every context (~T=F smaller, \[\]T=T, T&x=x, T\|x=T), so a primitive top moves only the exploitable class T; counts of unexploitable self-cooperators are unchanged at every size bound and for any atom set.

Prior art. Partial prior art.

Claim K53

In L with atoms p, q (guarded, size ≤ 14), 26449 machine classes (16756 behaviour classes) are unexploitable and self-cooperating; 387 of them behave as FairBot, and each other one gets a different outcome than FairBot against an explicit opponent.

Status: derived ů author check only.

Verified.

- `checks/K53-c046_qfull14.py`: guarded class counts 1..14 (279115 at 14); 249390 self-cooperators, 222941 exploited by re-solved explicit opponents; 26449 unexploitable (per-size list), 16756 behaviour classes; FairBot class 387 machine classes with per-size list; 26062 others separated from []p by re-solved opponents; rank-0 2468 (=K51 prefix); 1058 not CC with FB (smallest size 9), 15306 CC with FB and CB; literal evaluator cross-check — pass

Prior art. Partial prior art.

Claim K11

$X = \Box(p \vee (q \wedge \Box\bot))$ (size 7) has the same moves as FairBot $\Box p$ at every height against every modal opponent, since an opponent's height-0 move against any agent equals its move against DefectBot.

Status: derived ů author check only.

Verified.

- `checks/K11-c046_xfb_lean.py`: Lean 4 core theorem X_same_as_FairBot for every causal opponent operator (fixed points coincide at all heights), no sorryAx; 3 mutations rejected — pass (Lean-proved, no sorryAx)

Prior art. Possibly new: no prior art found in the owner's search.

Claim K18

For causal agents A, A' in height semantics: they make the same moves in every game against every causal opponent iff their DefectBot traces are equal and they agree on all realizable words ($p = q$ up to the first cooperation of A or its DefectBot trace).

Status: derived ů author check only.

Verified.

- `checks/K18-c046_realizable_lean.py`: Lean 4 core theorems game_exists, inputs_realizable, same_behaviour, separation, behaviour_iff, X_beheq_FB; SHA-pinned source; no sorryAx; 4 mutations rejected — pass (Lean-proved, no sorryAx)
- `checks/K18-c046_qfull12.py`: behav() realizable-word machines: 2574 classes, FairBot class has 86 machine classes with identical realizable-word machine to []p (consequence: FairBot behaviour against every opponent via same_behaviour) — pass

Evidence.

- reasoning: In any game the opponent's inputs are all-false until the agent or its DefectBot trace cooperates, so by causality its play equals its DefectBot play there; hence inputs are realizable, and by induction equal realizable behaviour gives equal games.

Prior art. Possibly new: no prior art found in the owner's search.

Claim K2

$A = \neg\Box\bot \wedge \Box(p \vee \Box\bot)$ (size 9) is unexploitable by every modal opponent, cooperates with its copy and with CooperateBot, yet mutually defects with FairBot; every rank-0 unexploitable self-cooperator of size ≤ 8 cooperates with FairBot.

Status: derived ů author check only.

Verified.

- `checks/K2-c046_ponly12.py`: A and ~[]p&[](p|[]F) are among the 591 unexploitable self-cooperators, A vs A = CC, A vs []p = DD, A vs ~F = CC; all such agents not cooperating with FB have size >= 9 — pass

Evidence.

- reasoning: Height semantics: A(0)=F, A(h)=AND of p(g) over 1\<=g\<h; self-play all C from height 1; vs FairBot FB(1)=A(0)=F forces DD from height 2; opponent tail D forces A tail D.

Prior art. Partial prior art.

Provenance. Run `20261020-1600`, claims K51, K53, K11, K18, K2, run-folder commit `e3d59e30bce37b58223f480d4f0d5a79bbff59ce`, published 2026-10-21.

3 Discussion

The counts depend on the agent language and the size measure; they say how common non-FairBot unexploitable self-cooperators are among small formulas, not in any absolute sense. "Unexploitable" is about what actually happens in play (truth in the standard model, as in Barasz et al.), not about what Peano arithmetic can prove about the agent; the LessWrong discussion points out that the two differ. Open questions: how the counts grow with size, and whether the non-FairBot classes fall into a small number of families.

4 How to verify

Each finding has its check scripts in `claims/only-one-fairbot-census/checks/` of the findings repository. Run each with `python -I <script>`; it passes if it exits with status 0. The checks are Python scripts and need the packages of the research image (the image's `requirements.txt`); the Lean checks also need `lean`.

5 Revisions

- 2026-10-21 — Production note: Lean proofs run inside check scripts are now described as such, separately from harness-built proofs

References

[1] Mihaly Barasz, Paul Christiano, Benja Fallenstein, Marcello Herreshoff, Patrick LaVictoire, and Eliezer Yudkowsky. Robust cooperation in the prisoner's dilemma: Program equilibrium via provability logic, 2014.

[2] Paul Beame and Søren Riis. More on the relative strength of counting principles. In *Proof Complexity and Feasible Arithmetics*, volume 39 of *DIMACS Series in Discrete Mathematics and Theoretical Computer Science*, pages 13–35. American Mathematical Society, 1998.

[3] Christoph Benzmüller. A (simplified) supreme being necessarily exists, says the computer: Computationally explored variants of gödel's ontological argument. In *Proceedings of KR 2020*, 2020.

[4] Christoph Benzmüller. A comment on modal collapse and ultrafilters in gödel's ontological argument. arXiv:2608.07578, 2026.

[5] Christoph Benzmüller. Gödel's and scott's variants of the ontological argument in lean 4 and tptp thf. arXiv:2609.26806, 2026.

[6] Christoph Benzmüller. Proofs without nominals: Gödel's ontological argument, its shallow embedding, and the open questions of the monatshefte notes. arXiv:2609.36279v4, 2026.

[7] Christoph Benzmüller and David Fuenmayor. Computer-supported analysis of positive properties, ultrafilters and modal collapse in variants of gödel's ontological argument. arXiv:1910.08955, 2019.

[8] Christoph Benzmüller and Lawrence C. Paulson. Quantified multimodal logics in simple type theory. *Logica Universalis*, 7(1), 2013.

[9] Christoph Benzmüller and Dana S. Scott. Notes on gödel's and scott's variants of the ontological argument. *Monatshefte für Mathematik*, 2025.

[10] Christoph Benzmüller, Leon Weber, and Bruno Woltzenlogel Paleo. Computer-assisted analysis of the anderson–hájek ontological controversy. *Logica Universalis*, 2017.

[11] Patrick Blackburn, Maarten de Rijke, and Yde Venema. *Modal Logic*. Cambridge University Press, 2001.

[12] George Boolos. *The Logic of Provability*. Cambridge University Press, 1993.

[13] Andrew Critch. A parametric, resource-bounded generalization of löb's theorem, and a robust cooperation criterion for open-source game theory. *Journal of Symbolic Logic*, 2019.

[14] Andrew Critch, Michael Dennis, and Stuart Russell. Cooperative and uncooperative institution designs: Surprises and problems in open-source game theory. arXiv:2208.07006, 2022.

[15] Kit Fine. Propositional quantifiers in modal logic. *Theoria*, 36(3), 1970.

[16] Melvin Fitting. *Types, Tableaus, and Gödel's God*. Kluwer, 2002.

[17] Murdoch J. Gabbay and Dominic P. Mulligan. Nominal Henkin semantics: simply-typed lambda-calculus models in nominal sets. arXiv:1111.0089, 2011.

[18] Gram_Stone. Modal chicken. LessWrong, 2015.

[19] Donald Hobson. Yet more modal combat. LessWrong / AI Alignment Forum, 2021.

[20] Wesley H. Holliday. A note on algebraic semantics for S5 with propositional quantifiers. *Notre Dame Journal of Formal Logic*, 60(2), 2019.

[21] Patrick LaVictoire, Benja Fallenstein, Eliezer Yudkowsky, Mihaly Barasz, Paul Christiano, and Marcello Herreshoff. Program equilibrium in the prisoner's dilemma via löb's theorem. In *AAAI Workshop on Multiagent Interaction without Prior Coordination*, 2014.

[22] Alex Mennen. Modal combat for games other than the prisoner's dilemma. LessWrong / AI Alignment Forum, 2017.

[23] Mladen Mika and Jakob Nordström. A generalized method for proving polynomial calculus degree lower bounds. In *30th Conference on Computational Complexity (CCC 2015)*, volume 33 of *LIPIcs*, pages 467–487, 2015.

[24] Reinhard Muskens. Intensional models for the theory of types. *Journal of Symbolic Logic*, 72(1), 2007.

[25] Caspar Oesterheld. A note on the compatibility of different robust program equilibria of the prisoner's dilemma, 2022.

[26] Alexander A. Razborov. Proof complexity of pigeonhole principles. In *Developments in Language Theory (DLT 2001)*, volume 2295 of *Lecture Notes in Computer Science*, pages 100–116. Springer, 2002.

[27] Søren Riis. *Independence in Bounded Arithmetic*. PhD thesis, University of Oxford, 1993.

[28] Søren Riis and Meera Sitharam. Non-constant degree lower bounds imply linear degree lower bounds. Technical Report TR97-048, Electronic Colloquium on Computational Complexity, 1997.

[29] Robert M. Solovay. Provability interpretations of modal logic. *Israel Journal of Mathematics*, 25:287–304, 1976.

[30] Aleksandar Todorov, Jesse ten Napel, and Alexander Müller. Parametric open source games. arXiv:2606.27068, 2026.

[31] transhumanist_atom_understander. Is there only one FairBot? LessWrong, with a comment by Zac Hatfield-Dodds, 2026.